AN INTRODUCTION TO HOMOLOGY

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Abstract. Homology groups are a fundamental invariant in topology. We shall build the theory of singular homology from scratch, while providing intuitive examples to motivate it along the way. We will roughly cover simplices, chain complexes, chain homotopies, relative homology and excision. We finish with Mayer-Vietoris sequences along with some applications.

Disclaimer: This is rushed and unfinished!!!

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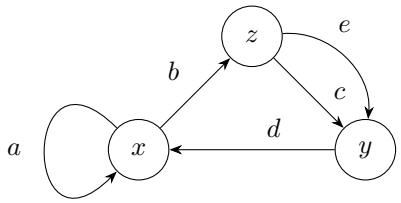
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1. Introduction

1.1. **Illustrating Examples.** We want to count the "1-dim holes" in the following graph X_1. One way is to consider all loops, for example $c \cdot e^{-1}, d \cdot a^3 \cdot b \cdot e$ where we are composing paths. But now, for a "hole" there are many paths corresponding to it due to the choice of basepoint, so we abelainise so that the paths don't depend on basepoint anymore. Define a $\mathbb{Z}$-formal sum of edges to be a 1-chain. We now ask the question, when is a chain a loop, or cycle? Observe that this happens when the number of times a cycle enters a vertex equals to the number of times it leaves that same vertex. Translating this to algebra, define $C_1(X_1)$ to be the free abelain group on the edges (1-chains) and $C_0(X_1)$ to be the free abelian group on the vertices (0-chains). These 1-chains inherently don't have any meaning but we can give 1-cycles some meaning since there must be some way of interpreting it as

Date: Revised November 5, 2026.

a loop in X_1.



Define the boundary map $\partial_1 : C_1(X_1) \to C_0(X_1)$ as

$$\partial_1 = \begin{bmatrix} 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & -1 & 0 & -1 \end{bmatrix}.$$

Obseve that the kernel of this map is precisely the 1-cycles. Indeed, $\ker \partial_1 = (a, b+c+d, c-e)$. Note that $\ker \partial_1 \cong \mathbb{Z}^3$, and the rank equal to the "1-dim holes" in X_1.

Construct the graph X_2 by "filling" in the part between c and e in X_1; call the filled in part A. (We are gluing a 2-cell). Then $c-e$ is still a cycle since $c-e \in \ker \partial_1$ but it not longer encloses a (1-)hole! Define the free abelain group $C_2(X_2)$ with generator A along with a boundary map $\partial_2(A) = c - e$. Since we don't want to consider the cycles that do not enclose a hole since they are the boundary of some 2-cell, we recitify our previous attempy by quotienting by $\operatorname{Im} \partial_2$, so define the "first homology group of X_1" as

$$H_1(X_2) = \frac{\ker \partial_1}{\operatorname{Im} \partial_2} \cong (a, b+c+d) \cong \mathbb{Z}^2$$

1.2. **Simplices.** The standard n-simplex is defined as

$$\Delta^n = \{(t_0, \ldots, t_n) \in \mathbb{R}^{n+1} \mid t_0, \ldots, t_n \geqslant 0, \sum_{i=0}^{n} t_i = 1\}.$$

This is the convex hull on vertices $(1, 0, \ldots, 0), (0, 1, \ldots, 0), \ldots, (0, 0, \ldots, 1)$. This will be a basic building block. This is a nice object, so we can define its face as $(0 \leqslant i \leqslant n)$,

$$\Gamma_i \Delta^n = \{(t_0, \ldots, t_n) \in \Delta^n \mid t_i = 0\}.$$

where Γ^i is the ith face operator. Indentify this with Δ^{n-1}. We can define the boundary of Δ^n as

$$\partial \Delta^n = \sum_{i=0}^{n} (-1)^i \Gamma_i \Delta^n.$$

For a general topological space X, define an n-singular simplex simply to be a continuous map $f : \Delta^n \to X$. It face is defined as

$$\Gamma^i \Delta^n \hookrightarrow \Delta^n \xrightarrow{f} X,$$

so $\Gamma^i f = f \mid \Gamma^i \Delta^n$. Similarly, it boundary is defined as

$$\partial f = \sum_{i=0}^{n} (-1)^i \Gamma^i f.$$

Define the n-singular chains as $\mathbb{Z}$ linear combinations of $f : \Delta^n \to X$, i.e. $\sum_i k_i f_i$. Call the free abelain group generated by *ALL* n-singular simplices to be $C_n(X)$, this is a massive group. The above bounday map gives a map $\partial_n : C_n(X) \to C_{n-1}(X)$.

Proposition 1.1. *The composition* $C_{n+1}(X) \overset{\partial_{n+1}}{\to} C_n(X) \overset{\partial_n}{\to} C_{n-1}(X)$ *is zero.*

Proof. Simply follow from the fact that

$$\Gamma_i \Gamma_j f = \begin{cases} \Gamma_{j-1}\Gamma_i f & j > i \\ \Gamma_j \Gamma_{i+1} f & \text{otherwise} \end{cases}.$$

□

The above proposition gives that $\partial^2 = 0$, hence $\operatorname{Im} \partial_{n+1} \subset \ker \partial_n$. We can define the nth-homology group of X as

$$H_n(X) = \frac{\ker \partial_n}{\operatorname{Im} \partial_{n+1}}.$$

This follows the same pattern as we saw in the examples, that is, cylces mod boundaries. We also have the following "complex"

$$\cdots \to C_{n+1}(X) \overset{\partial_{n+1}}{\to} C_n(X) \overset{\partial_n}{\to} C_{n-1}(X) \to \cdots$$

with $\partial^2 = 0$.

If the difference of two cycles is a boundary, then they are called *homologous*.

1.3. **Chain Complexes.** We abstract away some things from the above definition to define chain complexes. A chain complex $C_\bullet$ is a family of abelian groups $(C_n)_{n\in\mathbb{Z}}$ along with chain maps $\partial_n : C_n \to C_{n-1}$ for all $n \in \mathbb{Z}$ such that $\partial^2 = 0$. Define $Z_n = \ker \partial_n$ (n-cycles) and $B_n = \operatorname{Im} \partial_{n+1}$ (n-boundaries), then define $H_n = Z_n/B_n$, the nth-homology group of the complex $C_\bullet$.

We define a chain map $\varphi : C_\bullet \to D_\bullet$ as a family of maps $\varphi_n : C_n \to D_n$ such that the following commutes

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial} & C_n & \xrightarrow{\partial} & C_{n-1} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle \varphi_{n-1}} & & \downarrow{\scriptstyle \varphi_n} & & \downarrow{\scriptstyle \varphi_{n+1}} & & \\ \cdots & \longrightarrow & D_{n-1} & \xrightarrow{\partial} & D_n & \xrightarrow{\partial} & D_{n-1} & \longrightarrow & \cdots \end{array}$$

These chain complex and chain maps form a nice category (called an abelian category) that behaves just like R-mod. We shall call this category $Ch(Ab)$.

Observe that $\varphi : C_\bullet \to C'_\bullet$ gives us $\varphi_n(Z_n) \subset Z'_n$ and $\varphi_n(B_n) \subset B'_n$, this induces a map of homology, indeed $\varphi_* : H_n(C) \to H_n(C')$ given by $\bar{f} \mapsto \overline{\varphi f}$. This shows that $H_n : Ch(Ab) \to Ab$ is a functor.

For a continuous map $h : X \to Y$, we get an induces chain map $h_\# : C(X) \to C(Y)$ where $h_{\#n} : C_n(X) \to C_n(Y)$ is given by $h_\#(\sum_i k_i f_i) = \sum_i k_i (h_\# \circ f_i)$. This too induces maps to homology, indded $h_* : H_n(X) \to H_n(Y)$ given by $\bar{f} \mapsto \overline{hf}$.

1.3.1. *Chain Homotopies.* We now see how homology groups interact with homotopies of maps. First a definition:

Given $\varphi, \psi : C_\bullet \to C'_\bullet$, we define a chain homotopy as a family of maps $D_n : C_n \to C'_{n+1}$ such that

$$D_{n-1}\partial_n + \partial'_{n+1} D_n = \varphi_n - \psi_n.$$

(This diagram is NOT commutative)

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial} & C_n & \xrightarrow{\partial} & C_{n-1} & \longrightarrow & \cdots \\
 & & \Big\| \varphi_{n-1} & \swarrow^{D_n} & \Big\| \varphi_n & \swarrow^{D_{n+1}} & \Big\| \varphi_{n+1} & & \\
\cdots & \longrightarrow & C'_{n-1} & \xrightarrow{\partial} & C'_n & \xrightarrow{\partial} & C'_{n-1} & \longrightarrow & \cdots
\end{array}$$

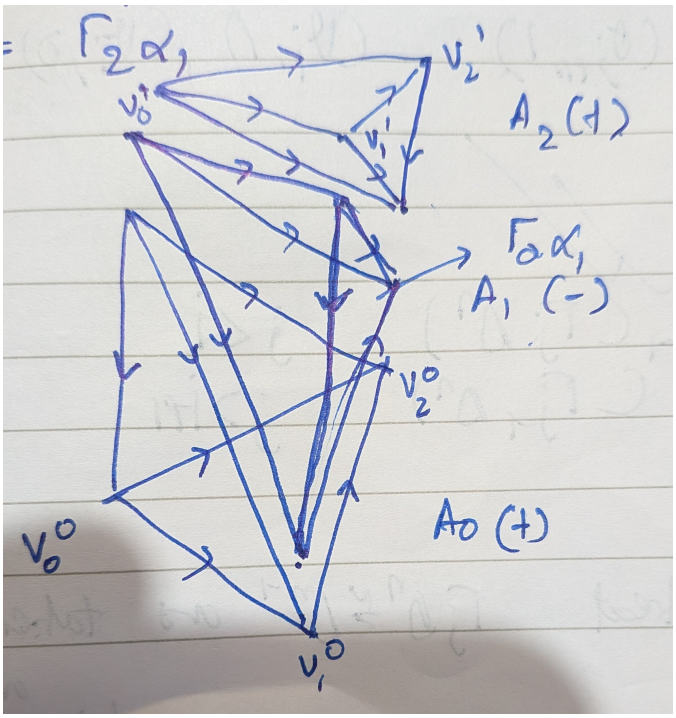
This might seem strange but we shall show that if $f, g : X \to Y$ such that $f \simeq g$, then $f_\#, g_\# : C(X) \to C(Y)$ are chain homotopic.

Proposition 1.2. *If $\varphi, \psi : C_\bullet \to C'_\bullet$ are chain homotopic then they induce the same map in homology, i.e. $\varphi_*, \psi_* : H_n(X) \to H_n(Y)$ agree.*

Proof. Calculate $\psi_n(c) - \phi_n(c)$ for $c \in \ker \partial_n \subset C_n$. □

We want to traingulate $\Delta^n \times I$ using $n+1$-singular simplices. Let Δ^n be the convex hull of $v_i = (0, \ldots, 1, \ldots, 0)$ where 1 is in the ith place. Let A_i be the convex hull of $(v_0, 1), \ldots, (v_i, 1), (v_i, 0), \ldots, (v_n, 0)$. Explicitly,

$$A_i = \{(t_0, \ldots, t_n) \times t \in \Delta^n \times I \mid t_0 + \cdots + t_{i-1} \leqslant t \leqslant t_0 + \cdots + t_i\}.$$



Define $\alpha_i(\Delta^n) : \Delta^{n+1} \to \Delta^n \times I$ which identifies Δ^{n+1} with A_i, these are $n+1$-singular simplices. Observe that $\Gamma^i \alpha_i = \Gamma^i \alpha_{i-1}$ for all $1 \leqslant i \leqslant n$ also $\Gamma^0 \alpha_0 = id_{\Delta^n} \times 0$ and $\Gamma^{n+1} \alpha_n = id_{\Delta^n} \times 1$. Let $\alpha(\Delta^n)$ be the chain

$$\alpha(\Delta^n) = \sum_{i=0}^{n+1} (-1)^i \alpha_i(\Delta^n),$$

this is precisely $\Delta^n \times I$. Using relations

$$\Gamma_j \alpha_i(\Delta^n) = \begin{cases} \alpha_{i-1}(\Gamma_j(\Delta^n)) & j < i \\ \alpha_i(\Gamma_{j-1}(\Delta^n)) & \text{otherwise.} \end{cases},$$

we get that

$$\partial\alpha(\Delta^n) = id_{\Delta^n} \times 0 - id_{\Delta^n} \times 1 - \alpha(\partial\Delta^n). \tag{1}$$

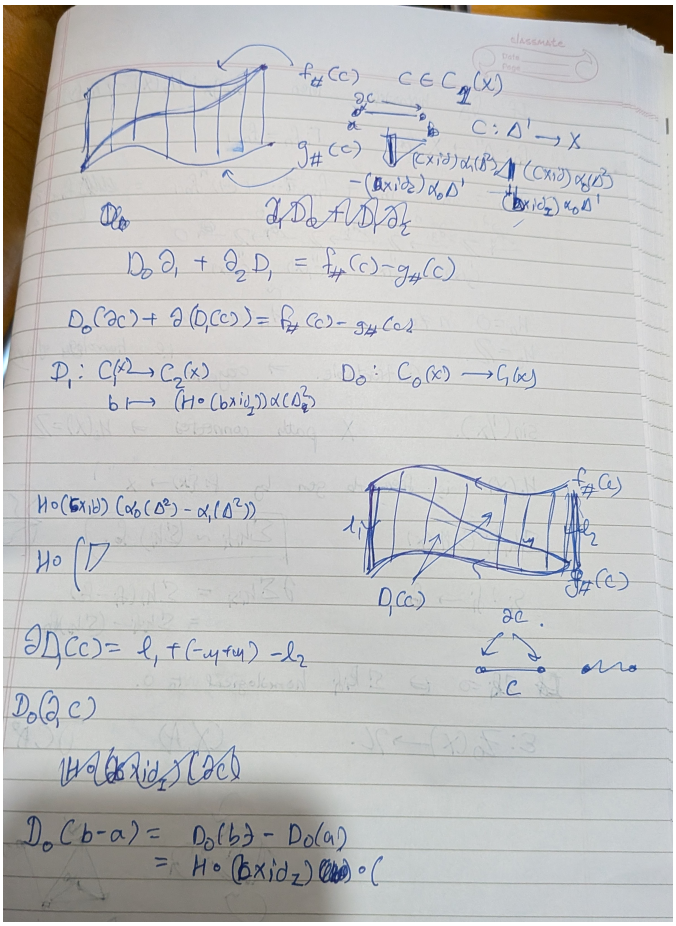
Let $f, g : X \to Y$ and let $H : X \times I \to Y$ be a homotopogy between f and g. Let $b \in C_n(X)$, i.e. $b : \Delta^n \to X$, then

$$\Delta^n \times I \overset{b \times id_I}{\to} X \times I \overset{H}{\to} Y$$

Observe that $\alpha(\Delta^n) \in C_{n+1}(\Delta^n \times I)$. Define $B = (H \circ (b \times id_I))_\# \alpha(\Delta^n)$, then $B \in C_{n+1}(Y)$. Call $D_n : C_n(X) \to C_{n+1}(Y)$ to be $b \mapsto B$, then (1) gives for $c \in C_n(X)$;

$$\partial D_n(c) + D_{n-1}(\partial c) = f_\#(c) - g_\#(c)$$

Corollary 1.3. *If $f, g : X \to Y$ are homotopy equivalent then $H_*(X) \cong H_*(Y)$.*



2. First Computations

2.1. **Quick detour: Reduced homology.** Given a simplicial complex $C(X)$, the reduced complex is $\widetilde{C(X)}$ with is the same as $C(X)$ but $\widetilde{C(X)}_{-1} = \mathbb{Z}$ and $\epsilon : \widetilde{C_0(X)} \to \mathbb{Z}$ is given by $f \mapsto 1$ for all 0-singular simplices. We have

$$H_n(X) = \begin{cases} \widetilde{H}_n(X) & n \neq 0 \\ \widetilde{H}_n(X) \oplus \mathbb{Z} & n = 0 \end{cases}$$

2.2. **Homology of a point.** Let $X = \{\text{pt}\}$For all n, there is a unique n-simplex $f_n : \Delta^n \to X$. So $\partial f_n = 0$ if n odd and f_{n-1} if n even. So we get

$$\cdots \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots$$

This show that $H_n(\text{pt}) = 0$ if $n \neq 0$ and $H_0(\text{pt}) = \mathbb{Z}$.

2.3. **Zeroth Homology of a path connected space.** If X is path connected then $H_0(X) = \mathbb{Z}$. It suffices to show that $\ker \epsilon = \operatorname{Im} \partial_1$. Clearly $\operatorname{Im} \partial_1 \subset \ker \epsilon$. Conversely, if $\sum_i k_i f_i \in \ker \epsilon$ then $\sum_i k_i = 0$. Fix a 0=chain $f_0 \in X$. Consider paths (1-simplices) $s_i : f_i \to f_0$, then $\partial_1(\sum_i k_i s_I) = \sum_i k_i(f_i - f_0) = \sum_i k_i f_i$.

2.4. **Homology and Path components.** Let $X = \cup_\alpha X_\alpha$, its path components. Since Δ^n is path connected for all n, the image of f (an n-simplex) must be contained in some X_α. So $C_n(X) = \oplus_\alpha C_n(X_\alpha)$. Similarly $Z_n(X)$ and $B_n(X)$ also split into a direct sum like this, hence

$$H_n(X) = \bigoplus_\alpha H_n(X_\alpha)$$

3. Zig Zag Lemma

Theorem 3.1. *if $0 \to A_\bullet \xrightarrow{f} B_\bullet \xrightarrow{g} C_\bullet \to 0$ is a short exact sequence in $Ch(Ab)$. Then we have a long exact sequence*

$$\cdots \xrightarrow{g_*} H_{n+1}(C) \xrightarrow{\delta} H_n(A) \xrightarrow{f_*} H_n(B) \xrightarrow{g_*} H_n(C) \xrightarrow{\delta} H_{n-1}(A) \xrightarrow{f_*} \cdots$$

Lemma 3.2. *(Snake Lemma) Given the following diagram.*

$$\begin{array}{ccccccc} & & A & \xrightarrow{u} & B & \xrightarrow{v} & C \longrightarrow 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h \\ 0 & \longrightarrow & A' & \xrightarrow{u'} & B' & \xrightarrow{v'} & C' \end{array}$$

there is a map $\delta : \ker h \to cokerf$ such that

$$\ker f \to \ker g \to \ker h \to cokerf \to cokerg \to cokerh$$

is exact everywhere.

Proof of Theorem. Apply snake lemma to

$$\begin{array}{ccccccc} & & \frac{A_n}{\partial A_n} & \longrightarrow & \frac{B_n}{\partial A_n} & \longrightarrow & \frac{C_n}{\partial A_n} \longrightarrow 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ 0 & \longrightarrow & Z_{n-1}(A) & \longrightarrow & Z_{n-1}(B) & \longrightarrow & Z_{n-1}(C) \end{array}$$

□

For $A \subset X$, apply the zigzag lemma to $0 \to C(A) \to C(X) \to C(X)/C(A) \to 0$. The homology groups of $C(X)/C(A)$ are called relative homology groups of X wrt A, denoted $H_n(X, A)$. We state without proof that if (X, A) is a good pair (i.e. A closed in X, there is an neighbourhood V of A st A is a deformation retract of V), then $H_*(X, A) \cong H_*(X/A)$.

Put $(X, A) = (D^n, S^{n-1})$, then $X/A = S^n$. Moreover $H_n(X) = 0$ since it contractable, using the long exact sequence above we get that $\widetilde{H}_i(S^n) \cong \widetilde{H}_{i-1}(S^{n-1})$. By induction on n,

$$H_k(S^n) = \delta_{kn}\mathbb{Z}.$$

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